

Engineering Mechanics: System of Forces, Resolution, and Resultant Force

High-Level Overview

In engineering mechanics, a **force** is an interaction that, when unopposed, changes or tends to change the state of motion or rest of a body. When multiple forces act simultaneously on a structure, machine, or particle, they form a **System of Forces**.

To analyze complex engineering systems, we must answer three fundamental questions:

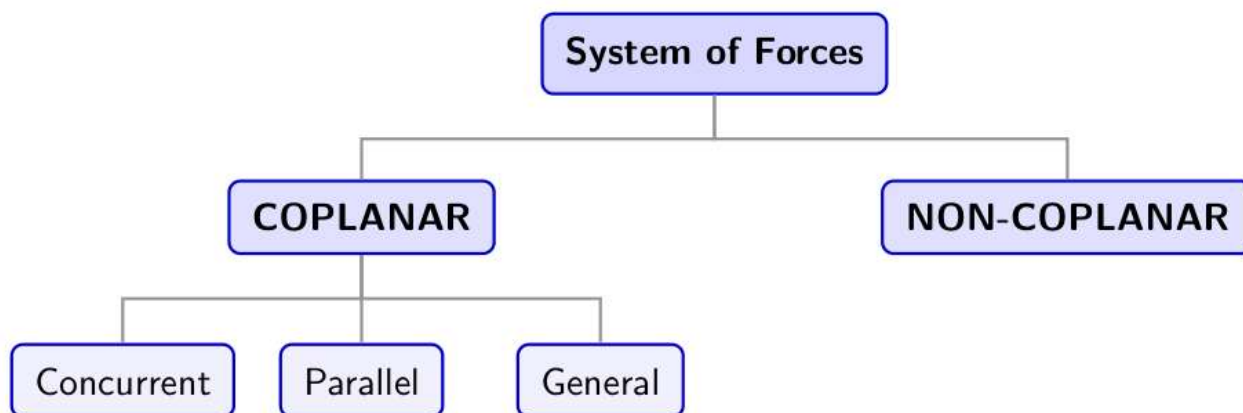
1. How are forces arranged relative to one another and the spatial dimensions (planes)?
2. How do we break down an inclined or angled force into simpler, orthogonal (perpendicular) directions?
3. How do we combine multiple arbitrary forces into a single equivalent force that produces the exact same physical effect?

Key Core Concepts

- **Force Vector:** A quantity defined by both a magnitude (how strong it is) and a direction (where it points).
- **Line of Action:** An imaginary infinite straight line extending along the direction vector of a force.
- **Coplanar System:** A set of forces operating strictly within a single two-dimensional plane.
- **Resolution of a Force:** The process of splitting a single inclined force vector into two perpendicular scalar components—typically along horizontal (x) and vertical (y) axes.
- **Resultant Force (R):** A single vector that completely replaces a system of multiple forces while producing the exact same dynamic or static effect on a body.

Systems of Forces Classification

Forces acting on a body are categorized based on their spatial orientation and whether their lines of action intersect or run parallel.



Visual Layout: A tree diagram branching from "System of Forces" at the top into two primary nodes: "COPLANAR" (left) and "NON-COPLANAR" (right). Beneath "COPLANAR", three child branches extend: "Concurrent", "Parallel", and "General".

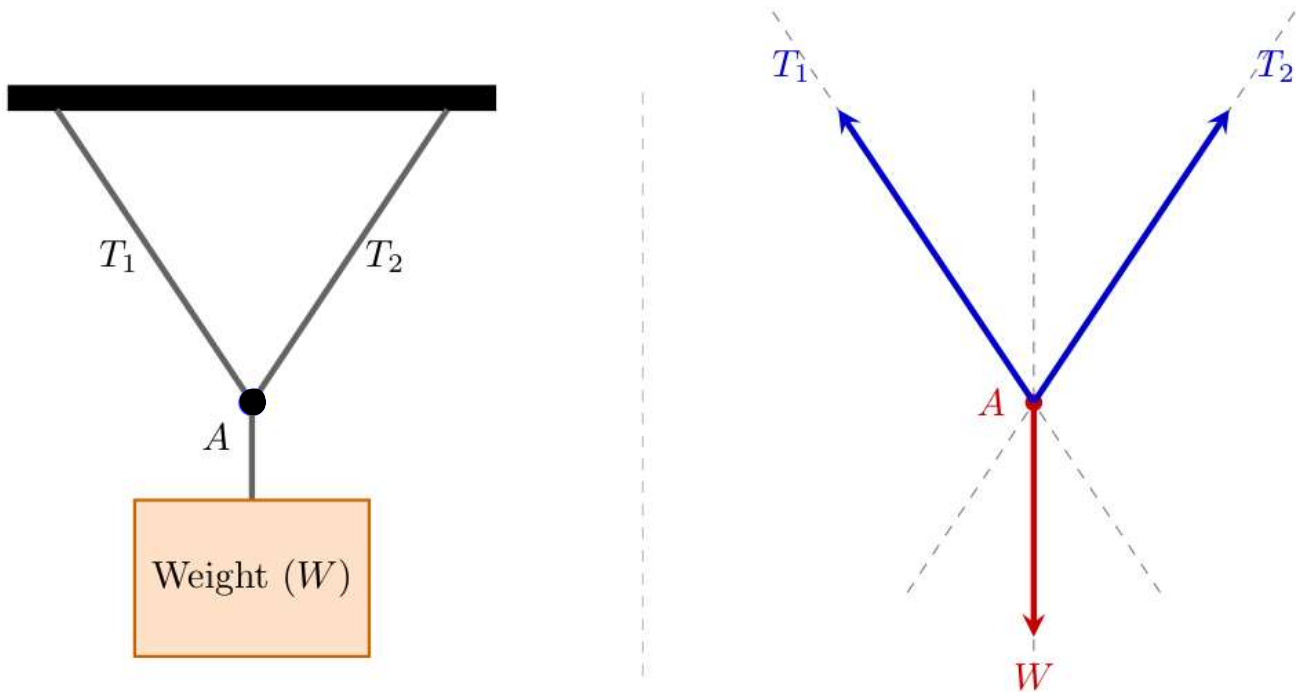
1. Coplanar vs. Non-Coplanar Systems

- **Coplanar System:** All forces in the system lie within a single, shared two-dimensional plane (e.g., xy -plane).
- **Non-Coplanar System:** Forces extend into three-dimensional space, lying across multiple, non-parallel planes.

2. Coplanar Classifications

A. Concurrent Force System

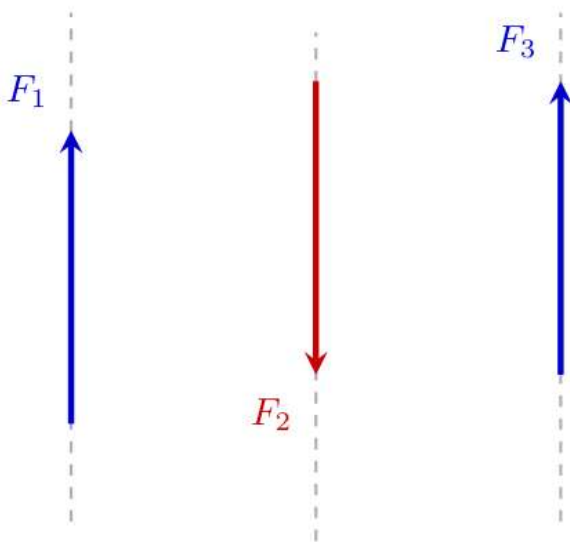
A coplanar system where the lines of action of **all** forces intersect at a single common point.



Visual Layout: A diagram illustrating a weight W hanging from a central point A , supported by two angled cables with tension forces T_1 and T_2 . Below it, a simplified vector diagram shows three force vectors (T_1 , T_2 , and W) radiating outwards from a single point of origin A .

B. Parallel Force System

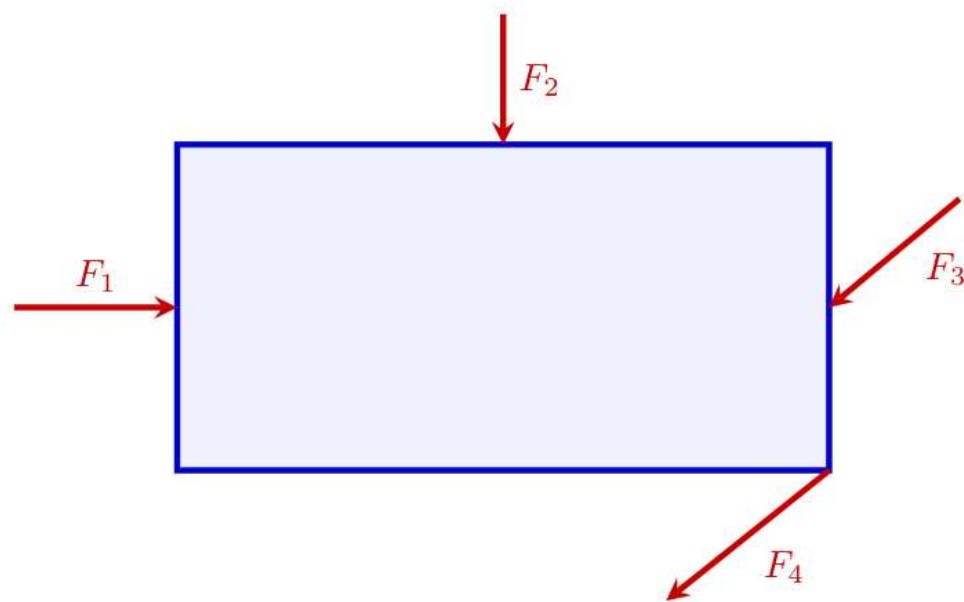
A coplanar system where the lines of action of all forces are parallel to one another and never intersect.



Visual Layout: A set of three vertical vectors pointing along parallel vertical lines. Two vectors point upwards and one points downwards, showing parallel lines of action.

C. General (Non-Concurrent, Non-Parallel) Force System

A coplanar system containing forces whose lines of action **do not** intersect at a single point and are **not** parallel to each other.



Visual Layout: A rectangular block subjected to four distinct forces: F_1 pointing horizontally to the right on the left face, F_2 pointing vertically downwards at the top center, F_3 pointing downwards and to the left on the right face, and F_4 pointing downwards and to the left from the bottom-right corner.

Summary Comparison of Force Systems

System Type	Shared Plane?	Lines of Action Intersect at a Single Point?	Lines of Action Parallel?
Coplanar Concurrent	Yes	Yes	No
Coplanar Parallel	Yes	No	Yes
Coplanar General	Yes	No	No
Non-Coplanar	No	Varies	Varies

Resolution of a Force

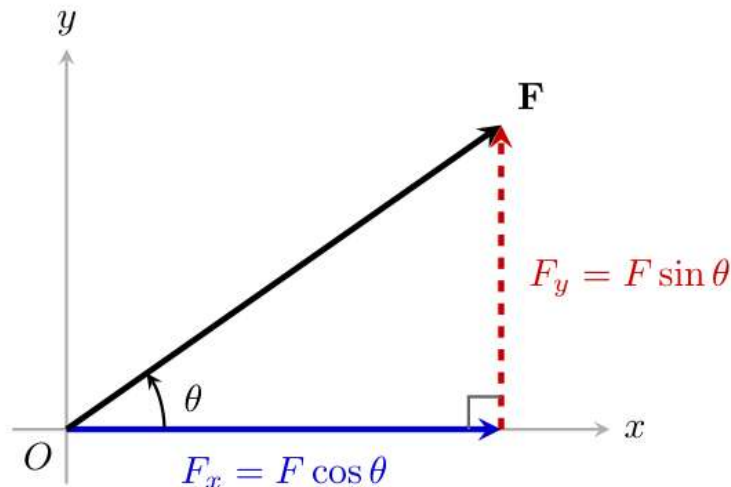
1. Fundamental Principle

Working with forces at arbitrary angles makes structural equilibrium calculations complex. **Resolution** is the mathematical procedure of decomposing a single vector force into two perpendicular (orthogonal) components:

- **Horizontal Component (F_x):** The projection of the force vector along the horizontal x -axis.
- **Vertical Component (F_y):** The projection of the force vector along the vertical y -axis.

2. Mathematical Formulation

Consider a force vector $\mathbf{F}$ of magnitude F acting at an angle θ measured directly relative to the **horizontal axis**.



Visual Layout: A right-angled triangle where the hypotenuse is a force vector $\mathbf{F}$ inclined at angle θ relative to the horizontal base. The horizontal base represents $F_x = F \cos \theta$, and the dashed vertical leg represents $F_y = F \sin \theta$.

Using standard right-triangle trigonometry:

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{F_x}{F} \implies F_x = F \cos \theta$$

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{F_y}{F} \implies F_y = F \sin \theta$$

Crucial Rule for Angle Measurement:

The component **adjacent** to the known angle θ always takes the **cosine (cos)** function, while the component **opposite** to the known angle takes the **sine (sin)** function.

- If angle θ is measured relative to the **horizontal axis**:

$$F_x = F \cos \theta, \quad F_y = F \sin \theta$$

- If angle ϕ is measured relative to the **vertical axis**:

$$F_x = F \sin \phi, \quad F_y = F \cos \phi$$

Step-by-Step Breakdown: Computing Resultant Force

To combine a system of multiple coplanar forces into a single equivalent force vector $\mathbf{R}$, follow this standardized four-step procedure:

Step 1: Sign Convention Establishment

Define standard Cartesian directions:

- **Horizontal (x-axis):** Forces pointing Right are Positive (+); Forces pointing Left are Negative (-);
- **Vertical (y-axis):** Forces pointing Upwards are Positive (+); Forces pointing Downwards are Negative (-).

Step 2: Summation of Components

Resolve every force vector into its respective horizontal and vertical projections, then algebraically sum them:

$$\Sigma F_x = \text{Sum of all horizontal force components}$$

$$\Sigma F_y = \text{Sum of all vertical force components}$$

Step 3: Resultant Magnitude

Combine the total scalar components using the Pythagorean Theorem:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

Step 4: Resultant Direction Angle

Determine the angle of inclination θ of the resultant force relative to the horizontal axis:

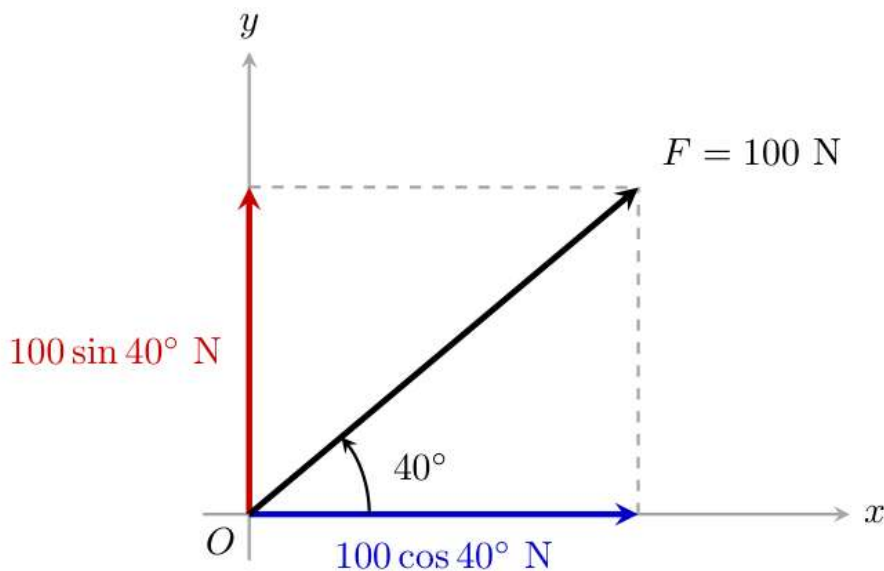
$$\theta = \tan^{-1} \left| \frac{\Sigma F_y}{\Sigma F_x} \right|$$

Worked Examples

Worked Example 1: Resolution of Basic Forces

Problem Statement:

Resolve a force magnitude $F = 100 \text{ N}$ acting upwards and to the right at an angle of 40° relative to the horizontal plane.



Visual Layout: A coordinate system showing a vector with magnitude 100 N inclined at 40° to the horizontal axis. Dashed vector projections show a vertical upward arrow labeled $100 \sin 40^\circ$ and a horizontal rightward arrow labeled $100 \cos 40^\circ$.

Detailed Solution:

1. Identify Given Data:

- Force Magnitude $F = 100 \text{ N}$
- Angle relative to horizontal $\theta = 40^\circ$

2. Calculate Horizontal Component (F_x):

$$F_x = F \cos \theta = 100 \cdot \cos(40^\circ)$$

$$F_x \approx 100 \cdot 0.7660 = 76.60 \text{ N} \quad (\rightarrow)$$

3. Calculate Vertical Component (F_y):

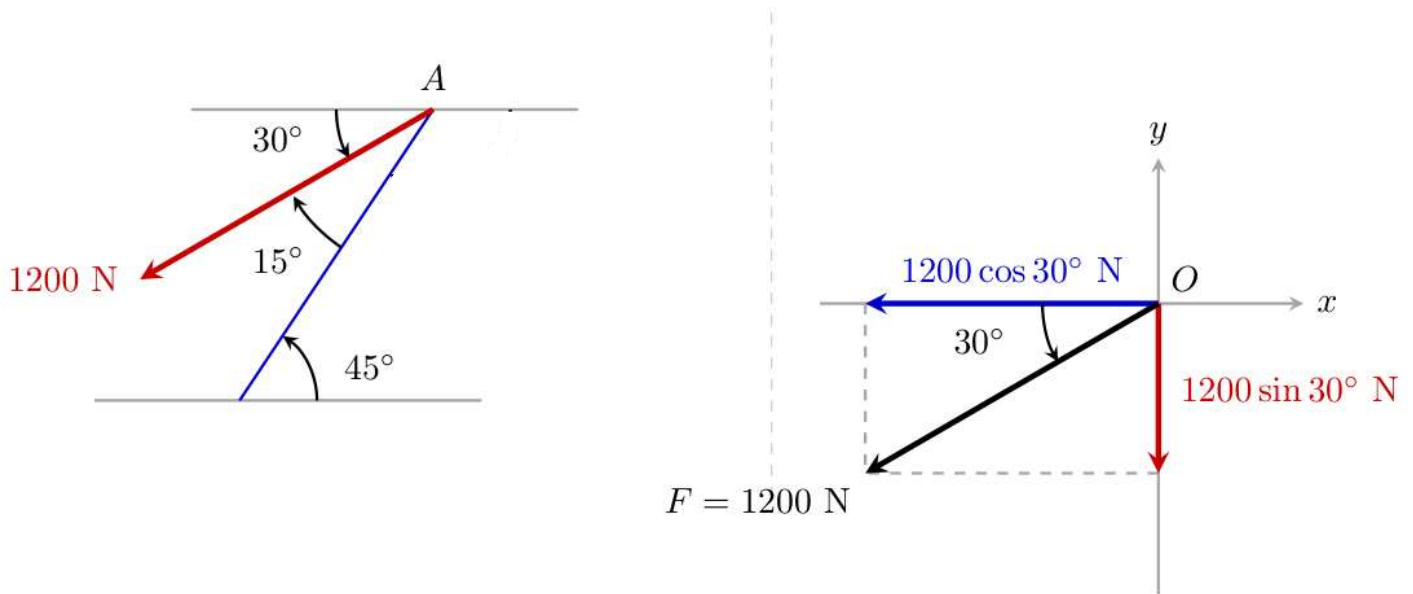
$$F_y = F \sin \theta = 100 \cdot \sin(40^\circ)$$

$$F_y \approx 100 \cdot 0.6428 = 64.28 \text{ N} \quad (\uparrow)$$

Worked Example 2: Determining Geometry and Angle via Alternate Interior Angles

Problem Statement:

A force vector of magnitude 1200 N acts downward and to the left along an inclined line. The incline makes an angle of 45° with the horizontal line below it. The line of action of the force is rotated further by 15° relative to that incline. Determine the true horizontal angle of inclination and resolve the force vector into horizontal and vertical components.



Visual Layout: A Z-shaped geometric reference figure demonstrating alternate interior angles of 45° . Beside it, a force line at an angle of 15° relative to a 45° inclined line is shown. The total angle relative to the top horizontal axis is deduced to be 30° ($45^\circ - 15^\circ$). A resolved coordinate system shows the 1200 N vector pointing into the third quadrant with components $1200 \cos 30^\circ$ pointing left and $1200 \sin 30^\circ$ pointing down.

Detailed Solution:

1. Determine the Effective Angle with the Horizontal:

- Using the Z-property of parallel lines (Alternate Interior Angles), the upper reference plane has an angle of 45° .
- Since the force arrow deviates towards the horizontal line by 15° , the angle formed by the force vector with the top horizontal reference line is:

$$\theta = 45^\circ - 15^\circ = 30^\circ$$

2. Determine Vector Directions (Quadrant Analysis):

- The force points downward and to the left (Quadrant III).
- Horizontal component F_x points **Left** (Negative Direction).
- Vertical component F_y points **Downwards** (Negative Direction).

3. Calculate Components:

- Horizontal Component (F_x):**

$$F_x = 1200 \cdot \cos(30^\circ) = 1200 \cdot 0.8660 = 1039.23 \text{ N} \quad (\leftarrow)$$

- Vertical Component (F_y):**

$$F_y = 1200 \cdot \sin(30^\circ) = 1200 \cdot 0.5 = 600.00 \text{ N} \quad (\downarrow)$$

4. Vector Representation:

$$\Sigma F_x = -1039.23 \text{ N}$$

$$\Sigma F_y = -600.00 \text{ N}$$